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1. NAME OF THE MEDICINAL PRODUCT

ALCON ATROPINE SULFATE 1% EYE DROPS

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

ALCON ATROPINE SULFATE 1% EYE DROPS: 1 ml of solution contains 10 mg atropine sulfate.
Preservative : 1 ml of solution contains 0.1 mg benzalkonium chloride.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Sterile, Eye Drops
A clear, colourless and viscous solution

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

ALCON ATROPINE SULFATE 1% EYE DROPS contains atropine sulfate, a parasympatholytic agent, which produces mydriasis and cycloplegia.

ALCON ATROPINE SULFATE 1% EYE DROPS is used for refraction or for dilation of the iris desired in acute inflammatory conditions of the iris and uveal tract.

4.2 Posology and method of administration

Posology

Use in adults

- For uveitis : 1 drop in the eye(s), 3 times daily.
- For refraction : 1 drop in the eye(s), repeated 1 hour before the examination.

Use in children

ALCON ATROPINE SULFATE 1% EYE DROPS is contraindicated in children less than 12 years because of the risk of serious systemic side effects (see sections 4.3, 4.4, 4.8 and 4.9).

Use in Elderly Population

Elderly patients may be higher risk for undiagnosed glaucoma as well as atropine-induced psychotic reactions and behavioural disturbances.

Use in patients with hepatic or renal impairment

The safety and efficacy of ALCON ATROPINE SULFATE 1% EYE DROPS have not been evaluated in patients with hepatic and renal failure.

Method of administration

For ocular use.

To prevent contamination of the dropper tip and solution, care must be taken not to touch the eyelids, surrounding areas or other surfaces with the dropper tip. Keep the bottle tightly closed when not in use.

Nasolacrimal occlusion or gently closing the eyelid after administration is recommended. This may reduce the systemic absorption of medicinal products administered via the ocular route and result in a decrease in systemic adverse reactions. This is particularly important in children.

After cap is removed, if tamper evident snap collar is loose, remove before using product.

If more than one topical ophthalmic product is being used, the products must be administered at least 5 minutes apart. Eye ointments should be administered last.

4.3 Contraindications

- Hypersensitivity to the active substance(s) or to any of the excipients.
- Hypersensitivity to belladonna alkaloids.
- Patients with known or suspected angle-closure glaucoma.
- Children less than 12 years (see section 4.4).
- Children with Down's syndrome, spastic paralysis or brain damage.

4.4 Special warnings and precautions for use

- Nasolacrimal occlusion or gently closing the eyelid after administration is recommended. This may reduce the systemic absorption of medicinal products administered via the ocular route and result in a decrease in systemic adverse reactions.
- Excessive use in children or certain individuals may produce systemic symptoms of atropine poisoning.
- ALCON ATROPINE SULFATE 1% EYE DROPS may cause increased intraocular pressure (see section 4.8).
  The possibility of undiagnosed glaucoma should be considered in some patients, such as elderly patients. Determine the intraocular pressure and an estimation of the depth of the angle of the anterior chamber prior to initiation of therapy to avoid glaucoma attacks.
- ALCON ATROPINE SULFATE 1% EYE DROPS -induced psychotic reactions and behavioural disturbances may occur in patients with increased susceptibility to anticholinergic drugs (see section 4.8). Use with caution in children and elderly patients, but reactions may occur at any age.
- Patients may experience sensitivity to light and should protect eyes in bright illumination.
- Because of risk of provoking hyperthermia (see section 4.8), use with caution in patients, especially in children, who may be exposed to elevated environmental temperatures or who are febrile.
- ALCON ATROPINE SULFATE 1% EYE DROPS contains benzalkonium chloride which may cause eye irritation and is known to discolour soft contact lenses. Avoid contact with soft contact lenses. Patients must be instructed to remove contact lenses prior to application of ALCON ATROPINE SULFATE 1% EYE DROPS and wait 15 minutes before reinsertion.
- **Paediatric population :**
  - Because of the risk of serious side effects, ALCON ATROPINE SULFATE 1% EYE DROPS is contraindicated in children below 12 years and caution is advised in older children, the lowest dose necessary to produce the desired effect should always be used. (See section 4.3, 4.4. 4.8 and 4.9).
  - Children, especially premature and low birth weight, or patients with Down syndrome, spastic paralysis or brain damage are particularly susceptible to central nervous system disturbances, cardiopulmonary and gastrointestinal toxicity from systemic absorption of atropine (See section 4.8).
  - Fair-skinned children with blue eyes may exhibit an increased response and/or increased susceptibility to adverse reactions.
  - Parents should be warned not to get this preparation in their children's mouth or cheeks and to wash their hands or cheeks following administration.

4.5 Interaction with other medicinal products and other forms of interaction

The effects of ALCON ATROPINE SULFATE 1% EYE DROPS may be enhanced by concomitant use of other drugs having antimuscarinic properties, such as amantadine, some antihistamines, phenothiazine antipsychotics, and tricyclic antidepressants.

4.6 Fertility, pregnancy and lactation

Fertility

Studies have not been performed to evaluate the effects of ocular administration of atropine or fertility.

Pregnancy

There are no or limited amount of data from the use of ALCON ATROPINE SULFATE 1% EYE DROPS in pregnant women. Animal studies are insufficient with respect to reproductive toxicity. There are documented systemic effects stemming from ophthalmic atropine use.

ALCON ATROPINE SULFATE 1% EYE DROPS is not recommended during pregnancy and in women of childbearing potential not using contraception.

Breast-feeding

It is unknown whether atropine is excreted in human milk after ocular administration. However, atropine and antimuscarinic agents have been shown to adversely affect lactation in preclinical and in clinical studies. A risk to the newborns or infants cannot be excluded.

ALCON ATROPINE SULFATE 1% EYE DROPS should not be used during breast-feeding.

4.7 Effects on ability to drive and use machines

Atropine may cause drowsiness, blurred vision and sensitivity to light. Patients receiving ALCON ATROPINE SULFATE 1% EYE DROPS should be advised not to drive or engage in other hazardous activities unless vision is clear.

4.8 Undesirable effects

The following adverse reactions have been identified from post-marketing surveillance following administration of ALCON ATROPINE SULFATE 1% EYE DROPS. Frequency cannot be estimated from the available data. Within each System Organ Class, adverse reactions are presented in order of decreasing seriousness.

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System Organ Classification	MedDRA Preferred Term (v12.1)
Immune system disorders	hypersensitivity
Psychiatric disorders	hallucination, confusional state, disorientation
Nervous system disorders	dizziness, headache
Eye disorders	eyelid oedema, photophobia, vision blurred, drug effect prolonged (mydriasis)
Cardiac disorders	tachycardia, bradycardia
Gastrointestinal disorders	intestinal obstruction, abdominal distension, vomiting
Skin and subcutaneous tissue disorders	erythema, rash
General disorders and administration site conditions	pyrexia

Description of selected adverse reactions
This drug produces reactions similar to those of other anticholinergic drugs. The central nervous system manifestations such as ataxia, incoherent speech, restlessness, hallucinations, hyperactivity, seizures, disorientation as to time and place, and failure to recognize people are possible. Other toxic manifestations of anticholinergic drugs are skin rash, abdominal distension in infants, unusual drowsiness, tachycardia, hyperpyrexia, vasodilation, urinary retention, diminished gastrointestinal motility, and decreased secretion in salivary and sweat glands, pharynx, bronchi and nasal passages. Severe reactions are manifested by hypotension with rapid progressive respiratory depression.

Symptoms of toxicity are usually transient (lasting a few hours), but may last up to 24 hours.

Mydriatics may increase intraocular pressure and provoke glaucoma attacks in patients predisposed to acute angle closure in particular geriatric patients (see section 4.4).

Prolonged use of mydriatics may produce local irritation characterized by conjunctivitis (follicular), ocular hyperaemia, eye oedema, eye discharge, and eczema.

Paediatric population
Use of ALCON ATROPINE SULFATE 1% EYE DROPS has been associated with psychotic reactions and behaviour changes in paediatric patients. Central nervous system reactions manifest similar to those listed above.

ALCON ATROPINE SULFATE 1% EYE DROPS can cause hyperpyrexia in children (see section 4.4).

Increased risk for systemic toxicity has been observed in children, especially premature and low birth weight, or patients with Down syndrome, spastic paralysis or brain damage with this class of drug (see section 4.4). Intestinal obstruction, abdominal distension and bradycardia were reported in premature or low birth weight infants.

4.9 Overdose
Systemic toxicity may occur following topical use, particularly in children. It is manifested by flushing and dryness of the skin (a rash may be present in children), blurred vision, a rapid and irregular pulse, fever, abdominal distension in infants, convulsions and hallucinations and the loss of neuromuscular coordination. Severe intoxication is characterized by central nervous system depression, coma, circulatory and respiratory failure, and death.

Treatment is symptomatic and supportive. In infants and small children the body surface must be kept moist.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties
Pharmacotherapeutic group : mydriatics and cycloplegics; anticholinergics, ATC code: S01FA01

Mechanism of Action
Atropine is an anticholinergic alkaloid that acts centrally and peripherally at the same time. In ophthalmology it is used as a cycloplegic and a mydriatic. It blocks the responses of the sphincter of the iris and the ciliary muscle to cholinergic stimulation, producing pupillary dilation (mydriasis) and paralysis of accommodation (cycloplegia).

Pharmacodynamics
It produces peak mydriasis within 40 minutes. The effect of atropine can last up to 2 weeks.

Clinical Studies
Atropine is an established medication.

Pediatric population
See Sections 4.2 and 4.3.

5.2 Pharmacokinetic properties

Absorption
Nonclinical data on absorption into ocular tissues with topical ocular administration are not available. The active enantionmer of atropine, l-hyoscyamine is absorbed systematically after topical ocular adminstration in man. The bioavailability (F) and time to maximum concentration (Tmax) after topical ocular administration were variable.

Distribution
Atropine is rapidly distributed throughout the body after intravenous administration which resulted in a volume of distribution greater than total body water. A biphasic disposition showed a distinctive distributional and elimination phases. Distributional properties differ between the two enantiomers, l-hyosyamide and d-hyosyamide. The observed plasma protein binding of atropine was low and variable (14-44%) from both in vitro and in vivo studies.

Biotransformation
Atropine is metabolized in the liver to two major metabolites, noratropine and N-atropine oxide along with the minor metabolites, tropine and tropic acid. The inactive enantiomer appears not to be metabolized.

Elimination
The elimination of atropine is similar after intraveneous or topical ocular administration with half-life range of 1.5-3.6 hours after ocular administration. Clearance values varied; however, the ocular route did not impact atropine's elimination. Up to 50% of atropine racemate is excreted unchanged in urine as the inactive enantiomer.

Linear/Non Linear Pharmacokinetics
The linearity of atropine's ocular pharmacokinetics has not been studied. Systemic pharmacokinetics of atropine was linear after intravenous administration from a dose of 1.4 mg to a dose of 2.2 mg.

Pharmacokinetics in Special Populations
The pharmacokinetics of atropine after topical ocular administration in pediatric populations and in patients with renal or hepatic insufficiencies has not been studied. Age-related differences in atropine pharmacokinetics after systemic dosing has been observed in younger pediatric (<2 years of age) and geriatric (>65 years of age) populations. The half-life was increased in both of these populations; with a decreased clearance found in elderly patients and increase volume of distribution in younger patients.

Pharmacokinetics and Pharmacodynamics relationship
The pharmacokinetic and pharmacodynamic relationship (PKPD) with topical ocular atropine has not been investigated; PKPD has been established with systemically administered atropine for systemic pharmacodynamic effects such as heart rate.

5.3 Preclinical safety data
Effects in non-clinical studies were observed only at exposures considered sufficiently in excess of the maximum human exposure indicating little relevance to clinical use in adults.

A low (5%) incidence of skeletal anomalies was observed when atropine was administered subcutaneously to pregnant mice at 50mg/kg; a dose greater than 700-fold higher than the maximal recommended adult dose on a mg/kg basis. Teratogenicity was not observed when atropine was given to pregnant rats or dogs.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients
Benzalkonium chloride, boric acid, hypromellose, sodium hydroxide and/or hydrochloric acid (to adjust pH) and purified water.

6.2 Incompatibilities
Not applicable.

6.3 Special precautions for storage
Do not store above 30°C

Discard one month after first opening.
Keep out of the sight and reach of children.

6.4 Nature and contents of container
ALCON ATROPINE SULFATE 1% EYE DROPS: 5 ml or 15 ml plastic DROPTAINER™ dispenser.

6.5 Special precautions for disposal
No special requirements.

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Date of revision: Nov 2024

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